

PET and Gestational Hypertension 09: Postpartum and Long-Term Risk

Companion reading handout for simultaneous listening.

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Let us start after delivery.

The baby is born. The placenta is out. The magnesium is running. The blood pressure is better. The team relaxes.

That relaxation can be dangerous.

Delivery removes the placenta, but it does not instantly normalize the endothelium, the kidneys, the liver, the brain, the lungs, or the vascular system. Pre-eclampsia can worsen postpartum. It can present for the first time postpartum. Eclampsia can occur postpartum. Stroke can occur postpartum.

The postpartum rule is simple.

Delivery is treatment. It is not discharge from risk.

Blood pressure often peaks around postpartum days three to seven. ISSHP links this to redistribution of extravascular fluid after delivery. NICE recommends blood pressure checks daily for the first two days and at least once between days three and five for gestational hypertension, with structured monitoring for pre-eclampsia. ISSHP recommends that women with antepartum hypertension have BP monitored at least once on days three to seven, when it is likely to be highest.

This timing matters because many patients are already home when the blood pressure peaks.

So discharge planning is not a formality. It is part of stroke prevention.

A strong discharge plan has concrete elements. It says when the next BP will be checked. It says what number should trigger a call or return. It says which symptoms require urgent assessment. It says which medication is being continued, started, reduced, or stopped. It says who is responsible for reviewing abnormal labs. And it says how the patient can actually access help on nights and weekends.

The vague instruction, come back if you feel unwell, is not enough for a disease that can present as headache, dyspnea, epigastric pain, confusion, or seizure after birth.

What symptoms matter postpartum?

The same dangerous symptoms matter, and sometimes more. Headache. Visual symptoms. Shortness of breath. Chest pain. Right upper quadrant or epigastric pain. Nausea and vomiting when concerning. Neurologic symptoms. Severe swelling with dyspnea. Decreased urine output. These symptoms should trigger assessment, not reassurance by the fact that the baby is already delivered.

Now postpartum antihypertensives.

NICE recommends starting treatment after birth for women not previously on medication if BP is at least one hundred fifty over one hundred. ISSHP recommends continuing antepartum therapy after birth and considering treatment for hypertension diagnosed before six days postpartum, targeting diastolic BP around eighty five. Local chronic hypertension guidance notes that BP may drop in the first days, then rise around days three to six.

Medication choices change after delivery because the fetus is no longer exposed through the placenta. Breastfeeding still matters, but many medications are compatible.

Common postpartum options include labetalol, nifedipine, hydralazine, and ACE inhibitors such as enalapril or captopril in many guidelines. Methyldopa is often avoided postpartum when alternatives exist because of concern about mood effects, though data are limited.

NICE is more specific. If methyldopa was used in pregnancy, stop it within two days after birth. Enalapril can be offered postpartum, with renal function and serum potassium monitored. If one drug is not enough, NICE suggests combinations such as nifedipine plus enalapril, and adding or substituting labetalol or atenolol when needed. It advises avoiding diuretics and angiotensin receptor blockers for treating hypertension in breastfeeding women when possible.

The mechanism matters. After birth, the renin-angiotensin system becomes a usable treatment target again, so ACE inhibitors can be very effective. But because they can affect renal function and potassium, they are not fire-and-forget prescriptions. They require context and follow-up.

Magnesium Postpartum

Now magnesium postpartum.

For pre-eclampsia with severe features, many protocols continue magnesium for twenty four hours after delivery or after the last seizure. Hadassah does this for PET with severe features during labor. The reason is that postpartum eclampsia risk persists even after the placenta is delivered.

Renal function still matters. If oliguria or creatinine rise occurs, magnesium can accumulate. Continue to monitor reflexes, respiratory rate, mental status, and urine

output.

NSAIDs

Now NSAIDs.

This is one of the places where guidelines differ. ACOG reviews reassuring evidence that ibuprofen and other NSAIDs do not meaningfully worsen postpartum blood pressure in patients with pre-eclampsia with severe features. ISSHP allows NSAIDs if other analgesics are ineffective and there is no acute kidney injury or risk factor for it. SOMANZ is more cautious and recommends against routine NSAID use in pre-eclampsia until more safety data are available, while allowing short inpatient use when alternatives are inadequate.

The practical interpretation is patient-specific. Avoid or monitor carefully if there is acute kidney injury, chronic kidney disease, uncontrolled hypertension, sepsis, postpartum hemorrhage, or fluid overload risk.

Labs and Renal Follow-Up

Now labs and renal follow-up.

NICE recommends postpartum labs at forty eight to seventy two hours for pre-eclampsia, and further follow-up based on abnormalities. At six to eight weeks, urine testing should assess persistent proteinuria. If proteinuria persists, reassess later and refer for kidney evaluation if abnormal. ISSHP recommends review at three months to confirm that BP, urinalysis, and lab abnormalities have normalized. If hypertension or proteinuria persists, investigate.

This is especially important for patients who may have had undiagnosed chronic kidney disease, chronic hypertension, or autoimmune disease before pregnancy.

The practical renal question is simple: did pregnancy reveal a temporary endothelial kidney injury, or did it uncover disease that was already there? Persistent proteinuria, persistent hypertension, or persistent creatinine abnormality after the postpartum recovery window should not be waved away as old pre-eclampsia. It may be chronic kidney disease, chronic hypertension, diabetes-related disease, autoimmune disease, or another renal diagnosis that needs long-term care.

Long-Term Risk

Now the long view.

Hypertensive disorders of pregnancy are not only pregnancy complications. They are cardiovascular risk markers.

Women with gestational hypertension or pre-eclampsia have increased later risk of chronic hypertension, cardiovascular disease, stroke, diabetes, venous thromboembolism, vascular dementia, and chronic kidney disease. ISSHP recommends counseling about these risks, lifetime cardiovascular risk framing rather than only ten-year risk, and annual medical review for the first five to ten years after hypertensive pregnancy.

This is not meant to frighten the patient. It is meant to convert a pregnancy complication into a prevention opportunity.

At follow-up, think about blood pressure, urine protein or albumin, kidney function, body mass index, glucose or diabetes risk, lipids, smoking, exercise, and primary-care connection. The postpartum visit should not end with, your BP is fine today, goodbye forever.

Recurrence and Future Pregnancy

Now recurrence.

ISSHP advises counseling that future gestational hypertension risk is at least four percent and future pre-eclampsia risk after pre-eclampsia is at least fifteen percent. The real risk depends on phenotype. Early-onset, severe, recurrent, or complicated disease can carry much higher recurrence risk.

NICE gives patients a more concrete way to hear this. After any hypertensive disorder of pregnancy, the chance of hypertension in a future pregnancy is roughly one in five. After pre-eclampsia, the chance of pre-eclampsia in a future pregnancy may be up to about one in six overall, higher if birth was needed earlier. If birth occurred between twenty eight and thirty four weeks because of pre-eclampsia, recurrence risk is roughly one in three. If birth occurred before twenty eight weeks, it approaches one in two.

Those numbers should be delivered carefully. They are not destiny. They are a reason for early booking, aspirin when indicated, baseline renal and proteinuria assessment when appropriate, and a surveillance plan that begins before the disease declares itself.

Future pregnancy planning should include early booking, baseline BP, baseline renal and proteinuria assessment when indicated, medication review, aspirin when indicated, calcium if intake is low, and optimization of chronic hypertension, diabetes, kidney disease, autoimmune disease, and weight.

Closing Synthesis

Let us close with the postpartum mental model.

The placenta is gone, but the vascular story is not over. Days three to seven are a danger window. Severe symptoms remain dangerous. Magnesium and antihypertensives may still be needed. Renal and proteinuria follow-up can reveal underlying disease. And the long-term cardiovascular risk is real.

Resident Reflex

The resident reflex is this: every patient with PET or GHTN needs a postpartum BP plan, symptom warnings, medication logic, renal follow-up when indicated, and long-term cardiovascular counseling. Delivery ends the pregnancy. It does not erase the disease history.